SIFT: Search Conjoint

Measuring Preferences and Search Costs in a Single Survey Instrument

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Conjoint analysis measures what consumers prefer among alternatives they have been shown. It is silent on what they would have looked at, what looking costs them, and what they never see at all. We introduce SIFT, a survey instrument in which the respondent faces a list-screen of alternatives, clicks through to inspect individual listings at will, and then chooses — a task that reproduces the two-stage list-then-detail architecture of the environments in which many category purchases are actually made. Each respondent completes a sequence of such tasks, and position, list-screen content, and detail-page content are randomized by design in every one. We fit the resulting click-and-choose data with a sequential search model in the tradition of Weitzman (1979), recovering both preference parameters, as conjoint does, and search costs, for which conjoint has no analogue. Two features distinguish the instrument from the structural search literature it borrows from. First, the within-respondent panel of search spells supports individual-level estimates of preferences and search costs jointly, where field applications typically observe a single spell per consumer and identify heterogeneity cross-sectionally. Second, the exogenous variation that field studies obtain only rarely — randomized position and information architecture — is designed in rather than found. The framework nests conjoint: as the search cost goes to zero every alternative is inspected, choice is made over fully realized utilities, and the model collapses to a standard choice-based conjoint model. Whether the search cost is distinguishable from zero is therefore a testable question rather than a maintained assumption, and the answer tells the analyst whether consideration or preference is the binding constraint in a category. We derive the likelihood, characterize what the designed variation buys for identification, and provide aggregate and hierarchical Bayesian estimation routines with replication code in R. A simulation study documents parameter recovery at commercial task counts, and an empirical application fielded in partnership with a consumer insights consultancy compares SIFT's preference estimates against conjoint part-worths from a split sample, tests whether estimated search costs are distinguishable from zero, and decomposes non-purchase into never-inspected and inspected-but-rejected — a distinction conjoint cannot draw, and one that carries

different managerial remedies.

1 Introduction

2 Related Literature

⚠ Citation verification status — remove before circulation

Every reference in this section was reconstructed from memory without database access and is **unverified**. Confidence flags follow each entry: **[H]** confident the paper exists and the details are roughly right; **[M]** confident the paper exists, unsure of journal, year, or coauthors; **[L]** believe something like this exists, may be conflating sources. Treat **[L]** items as search leads, not references. Nothing here should reach a referee, a proposal, or a client before it is checked against the record. The three entries marked **novelty gate** determine whether the paper's central claim survives; read those first.

2.1 Optimal Sequential Search

The theoretical backbone. The instrument's data-generating process is the Weitzman protocol run inside a survey.

- **Weitzman (1979), "Optimal Search for the Best Alternative,"** ***Econometrica.*** **[H]** The Pandora's box problem, and the model the instrument is built to estimate. Each box has a known prize distribution and a known opening cost. The optimal policy factors into three parts: a *reservation value* computed for each box independently of every other box; a *selection rule* (open boxes in descending reservation value); and a *stopping rule* (stop when the best realized prize exceeds the highest remaining reservation value). The independence of the reservation values is what makes the model estimable at survey scale.
- **Stigler (1961), "The Economics of Information,"** ***Journal of Political Economy.*** **[H]** The fixed-sample-size alternative: the consumer commits to n searches up front rather than deciding adaptively. The competing protocol, and the one SIFT's panel structure may be able to test against sequential search respondent by respondent.
- **McCall (1970),** ***Quarterly Journal of Economics.*** **[M]** Reservation-wage formulation of sequential search with recall; the labor-search ancestor of the marketing applications below.

Simultaneous versus sequential. These are two competing protocols, not two descriptions of one thing. Sequential search fits click-through-and-inspect; fixed-sample search fits "get three quotes." Honka and Chintagunta (2017) is the reference on distinguishing them empirically, and is directly relevant here because SIFT's within-respondent panel could run that test at the individual level rather than the population level.

2.2 Structural Search with Field and Clickstream Data

Context rather than precedent: all of this work is observational or platform data with no designed instrument. SIFT's credibility with an economics audience depends on connecting to it cleanly.

- **Hong and Shum (2006), *RAND Journal of Economics.*** [**H**] Recovers search costs from price distributions alone, without search data. The extreme case of how little the field has typically had to work with.
- **Hortacsu and Syverson (2004), *Quarterly Journal of Economics.*** [**H**] S&P 500 index funds. Search frictions with differentiated products, in a setting where the products are nearly identical and the frictions are not.
- **Kim et al. (2010), *Marketing Science.*** [**H**] Amazon view-rank data for camcorders. One of the first structural search models fit to online browsing data in marketing.
- **De los Santos et al. (2012), "Testing Models of Consumer Search Using Data on Web Browsing," *American Economic Review.*** [**H**] Comscore browsing data; tests sequential search against fixed-sample search and finds fixed-sample fits better in their setting. The empirical warning shot for assuming the Weitzman protocol rather than testing it.
- **Koulayev (2014), *RAND Journal of Economics.*** [**M**] Hotel search with click data; identification and estimation for differentiated products.
- **Honka (2014), *RAND Journal of Economics.*** [**H**] Auto insurance; separates search costs from switching costs. The template for arguing that an estimated friction is the friction you claim it is.
- **Honka and Chintagunta (2017), *Marketing Science.*** [**M**] Simultaneous or sequential; see above.
- **Chen and Yao (2017), *Management Science.*** [**M**] Sequential search with refinement — filters and sorts modeled as search actions — on Expedia clickstream data. Closest existing work in spirit to a list-screen interface, and the natural precedent if the instrument later adds filtering.
- **Ursu (2018), "The Power of Rankings," *Marketing Science.*** [**H**] Expedia's randomized-position experiment. The cleanest source of exogenous variation in the field literature, obtained once, as a rare gift from a platform. The contrast with SIFT is direct: here the same variation is free, designed in, and present in every task.
- **Ursu et al. (2020), *Marketing Science.*** [**M**] Search duration and time spent, rather than search incidence alone.
- **Yavorsky et al. (2021), *Quantitative Marketing and Economics.*** [**H**] Dealership visits in the U.S. auto market; identification of search costs from observed search sets. (Author's own; the estimation code is an implementation asset, see Section 5.)

- **Bronnenberg et al. (2016), "Zooming In on Choice," *Journal of Marketing Research.*** [M] Descriptive clickstream evidence on how consumers actually move through product pages. Face-validity evidence for the interface design in

    1.

- **2, *Quantitative Marketing and Economics.*** [M] — **novelty gate** Framework and consolidation review of the sequential search model. Best single entry point to the current state of model specification, identification, and estimation practice; read first to establish what is already settled.
- **Greminger (2022), *Management Science.*** [M] Search with discovery of alternatives — the consumer does not know the full choice set at the outset. Relevant to how much of the list-screen the instrument reveals up front.

2.3 Lab and Experimental Evidence on Search

Two experimental literatures that have barely spoken to each other, and the direct methodological precedent for running search as a designed task.

2.3.1 Does anyone follow the optimal rule?

- **SCHOTTER and BRAUNSTEIN (1981), *Economic Inquiry.*** [M] Early sequential-search experiments with real incentives.
- **Hey (1987), "Still Searching," *Journal of Economic Behavior and Organization.*** [M] Subjects use rules of thumb rather than optimal reservation strategies.
- **Cox and Oaxaca (1989).** [M] Finite-horizon search; broadly supportive of reservation-rule behavior with noise. The more favorable read.
- **Sonnemans (1998).** [M] Verbal-protocol evidence and classification of search strategies.
- **Brown et al. (2011).** [L] Real-time search in the laboratory and in the market.

Net read. Mixed, and this is the single largest scientific risk to the project. Behavior is directionally consistent with reservation-value logic, but the deviations are systematic rather than random: too little search relative to the optimum, order effects, and sensitivity to sunk costs. The paper should state this openly and early rather than let a referee find it.

2.3.2 Information acquisition: Mouselab and eye-tracking

Mechanically, the SIFT task is a Mouselab information board wearing a commercial skin.

- **Payne et al. (1988); Payne et al. (1993), *The Adaptive Decision Maker.*** [H] The Mouselab paradigm: information boards in which subjects click cells to reveal attribute values, with the acquisition sequence recorded. Descriptive and process-focused rather than structural.

- **Gabaix et al. (2006), "Costly Information Acquisition,"** ***American Economic Review.*** **[H] — novelty gate** A Mouselab-style experiment paired with a *structural* boundedly-rational model (directed cognition) estimated on the acquisition data. The closest existing template for the core maneuver — run an information-acquisition experiment, fit a structural search-type model to the acquisition record. Not Weitzman, but the same move.
- **Caplin et al. (2011), "Search and Satisficing,"** ***American Economic Review.*** **[H]** Choice-process data showing that people stop before finding the best available option. The satisficing alternative against which model fit should be reported.
- **Reutskaja et al. (2011),** ***American Economic Review.*** **[H]** Eye-tracking evidence on search under time pressure with supermarket-style displays.

Net read. The experimental apparatus is well validated and has been for decades. What has largely not been done is fitting a Weitzman model to acquisition data and reporting search costs in interpretable units.

2.4 Information Search inside Preference Measurement

The closest existing work in marketing, and the location of the sharpest threat to novelty.

- **Yang et al. (2015), "A Bounded Rationality Model of Information Search and Choice in Preference Measurement,"** ***Journal of Marketing Research.*** **[M] — novelty gate** Models within-task information search during a conjoint-style task using eye-tracking data, with an optimal-stopping flavor. If one paper threatens the contribution, it is this one. Read it first and read it carefully.
- **Yang et al. (2018),** ***Marketing Science.*** **[M]** Attention and information processing in incentive-aligned choice experiments.
- **Meißner et al. (2016),** ***Journal of Marketing Research.*** **[M]** Eye-tracking in conjoint; processing becomes measurably more efficient with practice. Directly relevant because a multi-task panel will exhibit learning across tasks, which appears in the model as drift in the search-cost parameter (Section 2.3, Section 9).

Distinction to preserve — and to verify before relying on it. These papers study search *within* a task, over the attributes of profiles already displayed, mostly in service of better preference estimates. SIFT studies search *across alternatives*, in a two-stage list-then-detail architecture, where the search cost is itself a deliverable rather than a nuisance parameter. Related, but not the same object. Confirm this holds before building on it.

2.5 Consideration Sets and Screening Rules in Conjoint

The existing marketing answer to "people do not evaluate everything," and the comparison a conjoint-literate reader will reach for first.

- **Gilbride and Allenby (2004), *Marketing Science.*** [**H**] A choice model with conjunctive, disjunctive, and compensatory screening rules. The incumbent formulation.
- **Hauser et al. (2010), *Journal of Marketing Research.*** [**H**] Disjunctions of conjunctions; cognitive simplicity in consideration rules.
- **Yee et al. (2007), *Marketing Science.*** [**M**] Greedoid-based noncompensatory inference.
- **Dzyabura and Hauser (2011), *Marketing Science.*** [**M**] Active machine learning for consideration heuristics.
- **Hauser (2014), *Journal of Business Research.*** [**M**] Review of consideration-set heuristics.

Sharpest one-line contrast. This literature *infers* screening rules from choice outcomes, treating consideration as latent structure. SIFT *observes* the consideration process directly and prices it in units of search cost. Inference versus observation.

2.6 Conjoint Practice, Incentive Alignment, and Commercial Precedent

- **Allenby et al. (2019), *Handbook of the Economics of Marketing.*** [**H**] Economic foundations of conjoint analysis; the statement of the workhorse model that SIFT nests.
- **Marshall and Bradlow (2002), *JASA.*** [**H**] A unified approach to conjoint analysis models; precedent for treating distinct response formats as one likelihood family.
- **Ding et al. (2005), *Journal of Marketing Research*; Ding (2007), *Journal of Marketing Research.*** [**H**] Incentive alignment in conjoint. If estimated search costs are to carry units that mean anything, the task needs real stakes; this is the precedent for how to supply them.
- **Sawtooth ACBC (adaptive choice-based conjoint).** Not an academic reference, but the closest commercial product to a two-stage consideration instrument: a screening section (must-haves, unacceptables) followed by a choice tournament. Study as design precedent, and as the thing a client may already have bought.
- **Virtual-shelf and shopper-simulation vendors.** Several field shelf-set exercises with click data, reported descriptively (time to first click, click share). No known case of backing a structural search model out of one. **This is where prior art most likely hides; check directly.**

2.7 Positioning

3 The SIFT Instrument

3.1 Task Architecture

A SIFT task presents the respondent with two kinds of screen, arranged the way online retail arranges them. The first is a **listing page**: a set of J alternatives displayed as an ordered vertical list, each occupying one row. The second is a **product page**, one per alternative, reached by clicking its row. The pair corresponds to what commercial practice calls the product listing page and the product detail page, and the correspondence is deliberate — the instrument is meant to look to the respondent like the environments in which the category is actually shopped.

The two screens differ in what they show. Every alternative is described by the same set of attributes, but the attributes are partitioned. A small number are displayed on the listing page for all J alternatives at once — position in the list, brand, and price are the natural candidates, and are the ones that list pages typically carry in practice. The remainder are shown only on an alternative's product page. A respondent who reads the listing page and clicks nothing therefore knows a little about every alternative; a respondent who opens a product page knows everything about that one alternative and still only a little about the rest.

This partition is what gives the task its structure. Listing-page attributes are free: the respondent acquires them by looking at the screen she is already on. Product-page attributes are costly: acquiring them requires a click, a page load, and the time to read what the page contains. The respondent chooses how many of those costs to pay and in what order.

The protocol within a task is as follows. The respondent begins on the listing page. She may open any alternative's product page, return to the listing page, and open another, as many times as she likes and in any order she likes. Every alternative she has already opened remains available to her — she may return to a previously viewed product page at no further cost, and she may select an alternative she viewed several clicks ago. The task ends in one of two ways. She selects an alternative to purchase, which she may do only from a product page she has opened, so that no alternative can be purchased sight-unseen. Or she declines to purchase anything and exits the task, which she may do at any point, including immediately, without opening a single product page.

Two features of this protocol deserve emphasis because the model in Section 4 rests on them. First, selection requires prior inspection, which is both realistic — one buys from the product page, not from the list — and what allows the observed choice to be read as a choice over resolved rather than expected utilities. Second, recall is free and unlimited, so a respondent is never punished for having looked at something and moved on. Together these place the task squarely in the Weitzman (1979) protocol: the listing page is a set of closed boxes whose contents are unknown but whose distributions are not, a click is the act of opening one at a cost, and the decision to stop and select is the decision to take the best prize already in hand rather than pay to open another. The no-purchase exit is the outside option, which the respondent may take whether or not she has searched.

The exit option matters for what the instrument can measure. A respondent who searches nothing and leaves, a respondent who searches half the list and leaves, and a respondent who searches half the list and buys are producing three different pieces of evidence about the same two parameters. Conjoint's no-choice option records only the first distinction between them.

Respondents complete a sequence of tasks rather than one, exactly as in choice-based conjoint. Each task is a fresh listing page with a fresh set of alternatives, and the respondent searches and chooses within it independently. The current design fields ten tasks per respondent. This is the feature that separates the instrument most sharply from the field literature it borrows from. A platform dataset typically observes one search spell per consumer, so preferences and search costs must be told apart across consumers. Here each respondent supplies a panel of search spells, and the same respondent's behavior across tasks can be used to separate what she likes from what looking costs her.

3.2 Randomization and Experimental Design

Three things vary from task to task, and all three vary by design rather than by circumstance.

Position. The order in which the J alternatives appear on the listing page is randomized independently of everything else about them. An alternative that appears first in one task appears fourth in another, with the same attributes and the same competitors. Position is thus uncorrelated by construction with quality, price, brand, and every other attribute — the correlation that makes observational ranking data so difficult to use, since platforms rank on relevance and relevance is a function of exactly the things that drive utility. Field work has obtained this variation only rarely and only as a gift, most cleanly in Ursu (2018). Here it is free and present in every task.

Composition. The set of alternatives available on the listing page varies across tasks: which alternatives appear, and how many. This is the analogue of varying the choice set in a conjoint design, and it serves the same purpose of placing each alternative against varied competition, but it does additional work here because the attractiveness of the alternatives a respondent has *not* yet opened is what determines whether she keeps searching.

Attribute levels. The attributes themselves are fixed across tasks — the same characteristics describe every alternative in every task, as in conjoint — while the levels those attributes take on are varied across alternatives and across tasks according to an experimental design. This is conventional conjoint practice and needs no defense; what is new is only that some of these attributes are revealed on the listing page and the rest behind a click.

A fourth lever is available and is not yet part of the design: the *partition* itself — which attributes appear on the listing page and which are held back for the product page — could be randomized across respondents rather than fixed. Doing so would make the information architecture an experimental treatment, and would let the instrument speak to what belongs on a list page versus a detail page. It also complicates identification, since it changes what the respondent knows before searching. We flag it here as a design option and return to it in Section 8.

We defer the question of what each of these three sources of variation identifies to Section 5, and the question of how they enter the likelihood to Section 4. The point to carry forward from this section is only that the variation is designed rather than found: it is orthogonal by construction, it is present in every task rather than in a single natural experiment, and it costs nothing to produce.

3.3 Data Recorded

The instrument records the full interaction, not merely its outcome. For each respondent i and task t we retain three layers.

The design. The complete attribute matrix for all J alternatives shown, whether or not the respondent ever saw them; the position each alternative occupied on the listing page; and the partition of attributes into listing-page and product-page sets. The design is known without error because it was generated, which is a meaningful advantage over field data, where the analyst typically knows what was clicked but must reconstruct what was displayed.

The event stream. Every action the respondent takes, in order, with a timestamp: each product-page opening (which alternative, at which position, at what time), each return to the listing page, each re-opening of a previously viewed product page, and the terminal event — a purchase of a named alternative or an exit without purchase. The event stream is the raw object; everything below is derived from it.

Derived search quantities. From the event stream we construct the objects the model in Section 4 takes as data: the *search set*, meaning which alternatives were opened; the *search order*, meaning the sequence in which they were opened; the *stopping point*, meaning how many alternatives were opened before the respondent stopped; and the *choice*, meaning which member of the search set was purchased, or that none was. Timing data yield a parallel set of duration measures — time to first click, dwell time per product page, total task duration — which are informative about search effort in the sense of Ursu et al. (2020) and which will also be needed to net out any imposed latency from voluntary reading time.

It is worth stating plainly what this means for the alternatives a respondent never opened. They are not missing data. Under the model they are the outcome of a decision — the respondent judged that opening them was not worth the cost, given what she had already found — and they carry information about both her preferences and her search cost. Non-inspection is the observation that conjoint has no way to record, and recovering it is much of the point of the instrument.

4 Model

4.1 Setup and Notation

We follow the notation of 2 wherever the two are compatible, so that readers arriving from the sequential-search literature can map our objects onto theirs without translation. Two departures are forced on us by the instrument, and one by the object being searched over; we flag all three below. Section 11.1 gives a crosswalk to 2 and Yavorsky et al. (2021) for readers coming from either.

Indices. Respondent $i = 1, \ldots, N$ completes tasks $t = 1, \ldots, T$. Task t presents respondent i with a listing page holding the alternatives $\mathcal{J}_{it} = \{1, \ldots, J_{it}\}$, together with the no-purchase option $j = 0$, which is always available and never requires a click. The index j names an *alternative*, not its rank and not its screen position: because position is randomized (Section 3.2), the paper must hold three things apart at once — an alternative's identity j, the position it occupied on the listing page in task t, and its rank in the respondent's reservation-value ordering. Search order is carried by a separate index h, where $h = 1$ denotes the first product page the respondent opened.

Utility. Respondent i's utility from alternative j in task t is

$$u_{itj} \;=\; \underbrace{\mathbf{X}'_{itj}\boldsymbol{\beta}_i + \mu_{itj}}_{\delta_{itj}\text{ (known before clicking)}} \;+\; \underbrace{\mathbf{L}'_{itj}\boldsymbol{\kappa}_i + \varepsilon_{itj}}_{\text{(revealed by clicking)}}\;, \tag{1}$$

where $\mathbf{X}_{itj}$ collects the attributes displayed on the listing page, $\mathbf{L}_{itj}$ the attributes displayed only on the product page, μ_{itj} is a pre-search taste shock observed by the respondent but not by us, and ε_{itj} is a post-search taste shock. Write $\xi_{itj} = \mathbf{X}'_{itj}\boldsymbol{\beta}_i$ for the observed part of pre-search utility and $\delta_{itj} = \xi_{itj} + \mu_{itj}$ for pre-search utility. Both shocks are mean-zero normal, $\mu_{itj} \overset{iid}{\sim} N(0, \sigma^2_\mu)$ and $\varepsilon_{itj} \overset{iid}{\sim} N(0, \sigma^2_\varepsilon)$, with $\sigma_\mu = 1$ imposed as the scale normalization. The no-purchase option has $u_{it0} = \mathbf{X}'_{it0}\boldsymbol{\beta}_i + \mu_{it0}$ with no post-search component: nothing about it is learned by clicking, so its utility is resolved from the outset.

The term $\mathbf{L}'_{itj}\boldsymbol{\kappa}_i$ is the departure that matters most. In the match-value formulations that dominate empirical work, everything learned through search is a scalar shock ε_{itj} unobserved by the researcher both before and after search. Here the product page reveals *attributes we designed*, so part of what search delivers is observed, priced by $\boldsymbol{\kappa}_i$, and — critically — drawn from a distribution the analyst chose rather than assumed. 2 give exactly this general form in their §2.4 and note it is uncommon in practice; SIFT is a case where it is unavoidable, since recovering $\boldsymbol{\kappa}_i$ for product-page attributes is half of what the instrument is for. The reservation-value discussion below takes up what this costs computationally.

Search costs. Opening alternative j's product page costs

$$c_{itj} \;=\; \exp\!\Big(\mathbf{Z}'_{itj}\boldsymbol{\gamma}_i\Big)\,, \tag{2}$$

exponentiated to enforce positivity, as is standard. $\mathbf{Z}_{itj}$ holds a respondent-level intercept and the randomized listing-page position, and may hold listing-page attributes that plausibly shift salience. Overlap between $\mathbf{X}$ and $\mathbf{Z}$ is permitted; Section 5 takes up what separates the two.

Where the randomness lives. Reservation values must carry a stochastic component, or search order would be a deterministic function of observables and no observed order could be rationalized (Ursu et al. 2024). Two specifications supply one. The first, used above and by most of the literature, puts a pre-search taste shock μ_{itj} in utility. The second, due to Chung et al. (2024), drops μ_{itj} and instead treats the search cost itself as random at the respondent-alternative level, with c_{itj} drawn from a distribution whose location carries the position effect. The two are observationally similar but econometrically quite different: under the first, the scale of reservation utilities and the scale of realized utilities are jointly determined by σ_μ and σ_ε, and estimating both requires an auxiliary step; under the second, search-cost dispersion fixes the scale of reservation values while σ_ε alone fixes the scale of realized utilities, and everything is estimable in one pass.

2 §4.4.1 object to the second on interpretive grounds, asking why the cost of clicking a product detail page would vary at the consumer-product level. The objection has least force here, because we randomize the thing that makes it vary: the same alternative sits at a different position for different respondents and in different tasks, so respondent-alternative variation in click cost is designed in rather than assumed. Section 5 takes up which specification to estimate; Chung et al. (2024) show that relative preference parameters survive getting this choice wrong, though search-cost *levels* do not.

Search and choice. Let $\mathcal{S}_{it} \subseteq \mathcal{J}_{it}$ denote the set of alternatives whose product pages respondent i opened in task t, $\bar{\mathcal{S}}_{it}$ its complement, $H_{it} = |\mathcal{S}_{it}|$ the number of clicks, and $y_{it} \in \mathcal{S}_{it} \cup \{0\}$ the alternative purchased, with $y_{it} = 0$ recording exit without purchase. Recall is free, so y_{it} may be any member of $\mathcal{S}_{it}$, and selection requires prior inspection, so it may not be drawn from $\bar{\mathcal{S}}_{it}$.

Reservation values. The reservation value z_{itj} equates the marginal cost of opening a product page to the marginal expected benefit,

$$\int_{z_{itj}}^{\infty} (u - z_{itj})\, dF_{itj}(u) \;=\; c_{itj}, \tag{3}$$

where F_{itj} is the respondent's belief about u_{itj} prior to clicking. Additive separability gives $z_{itj} = \delta_{itj} + g(c_{itj})$, with g decreasing in search cost and depending on the belief distribution only through the post-search component.

Here Equation 1 exacts a price, though a smaller one than it first appears. When the post-search component is a single normal shock, g has the familiar closed form $g(c) = m(c/\sigma_\varepsilon)\,\sigma_\varepsilon$ with

m solving $c/\sigma_\varepsilon = \phi(m) + m[\Phi(m) - 1]$ (Kim et al. 2010), and this is what every existing implementation computes. With designed product-page attributes the pre-click belief about $\mathbf{L}'_{itj}\boldsymbol{\kappa}_i + \varepsilon_{itj}$ is instead a *mixture* of normals — one component per attribute-level combination the design can produce, with weights fixed by the design — and g has no scalar closed form.

Two properties of a designed instrument keep this cheap. First, the mixture is *common across alternatives*: a balanced design draws $\mathbf{L}_{itj}$ from the same distribution at every position, so the respondent's pre-click belief is identical for every alternative she has not yet opened, and the mixture enters the model only through g rather than alternative by alternative. The same balance delivers the independence across j that Weitzman's rules require. Second, the left side of Equation 3 is then a probability-weighted sum of normal partial expectations, strictly decreasing in z_{itj}, so one root-find recovers g for each distinct search cost — and because position is categorical, the number of distinct search costs per respondent is the number of positions, not the number of alternative-task pairs. The scalar-normal case is the one-component special case, and nothing in Weitzman's rules requires normality, so the decision rules of Section 4.2 carry over unchanged.

A variance-matched normal approximation is also available, and it is the specification we estimate. Absorbing the design mean of $\mathbf{L}'\boldsymbol{\kappa}_i$ into δ_{itj} and replacing the mixture with a normal of variance $\tilde{\sigma}_i^2 = \boldsymbol{\kappa}_i'\mathrm{Var}(\mathbf{L})\boldsymbol{\kappa}_i + \sigma_\varepsilon^2$ returns the model exactly to the Kim closed form with an inflated post-search standard deviation. This is more than a computational convenience. The benefit of search — the object the literature normalizes away for want of variation to estimate it (Ursu 2018; Morozov et al. 2021) — is here *partly known*, because $\mathrm{Var}(\mathbf{L})$ is fixed by the design rather than assumed. Because g then depends on its arguments only through $c/\tilde{\sigma}_i$, a single one-dimensional inverse serves every respondent, which is what keeps estimation tractable at panel scale (Section 5.5). We report the exact mixture as a generalization and use it as a robustness check.

Parameters. Collect the respondent-level parameters in $\boldsymbol{\theta}_i = (\boldsymbol{\beta}_i, \boldsymbol{\kappa}_i, \boldsymbol{\gamma}_i)$ and the population parameters governing their distribution in $\boldsymbol{\Omega}$. We reserve $\boldsymbol{\theta}$ for the parameter vector throughout and write $\log \sigma_\varepsilon$ explicitly where the log scale is meant.

4.2 Optimal Search Policy

4.3 Likelihood

4.4 Conjoint as the Zero-Search-Cost Special Case

4.5 Alternative Search Protocols

5 Identification and Estimation

Search models are identified from patterns that ordinary choice data cannot produce: *which* alternatives a respondent opened, *in what order*, *how many*, and *which* of the opened alternatives was ultimately bought. This section sets out what each of those margins identifies, what SIFT's designed variation adds to arguments that field studies must make on observational grounds, and what remains normalized rather than estimated.

Throughout, the object of interest is the pair $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ — preferences and search costs — together with the post-search standard deviation $\tilde{\sigma}$, which most of the literature fixes to one and which we argue must be estimated.

5.1 What the Designed Variation Buys

5.1.1 Position enters search costs, not utility

The identifying assumption underlying every position-based search-cost shifter is that a listing's position affects what it costs to inspect an alternative, and nothing else. Ursu (2018) §4.3 takes this seriously rather than asserting it, enumerating the ways position could instead enter the model — through pre-search utility, through the post-search spread, or through combinations — and ruling them out with observational tests. Because SIFT randomizes position within every task and records the full click sequence, we can run those tests as experimental contrasts rather than conditional comparisons.

Two statistics suffice, and neither requires fitting a model.

T1. Among clicked alternatives, does position predict purchase? T2. Is the first click near the top of the page?

> **!** The sign of T1 is the opposite of the natural intuition
>
> It is tempting to reason that if position acts only on costs then it should have no bearing on what gets bought once someone has looked, so a flat T1 indicates position-in-cost. Simulation says otherwise, and the reason is selection. When position raises the search

cost, an alternative in a *bad* position must have unusually high pre-search utility to be worth opening at all. Conditional on having been clicked, bad-position alternatives are therefore *better* than good-position ones, and T1 comes out strongly **positive**. When position instead enters pre-search utility, the direct effect and the selection effect roughly cancel and T1 is approximately **flat**.
In our simulations (400 respondents $\times$ 8 tasks) T1 has slope $+0.39$ ($p \approx 5 \times 10^{-15}$) when position acts on costs, and -0.05 ($p = 0.06$) when it acts on utility. Reading a flat T1 as evidence for position-in-cost would get the answer exactly backwards.

T2 separates the remaining case. If position widens the post-search spread, the reservation value *rises* with position, so the worst-placed alternatives would be opened first. That is what we see: mean first-click position 4.96 against a uniform benchmark of 3.5, versus 2.21 and 2.57 in the cost and utility arms.

The two tests together form a decision rule:

T2	T1	position acts on
first click near top	strongly positive	**search cost**
first click near top	flat	pre-search utility
first click near bottom	—	post-search spread

We regard this as a design check to be run and reported in every SIFT study rather than assumed. Details, code, and the full three-arm simulation are in the position comparison.

5.1.2 Separating preferences from search costs

The sharper problem is not whether position shifts costs but whether a high search cost can be distinguished from a low taste. Morozov et al. (2021) gives the cleanest statement of the separation (his §4.3.2): stopping decisions depend on the *realized* utilities of alternatives already searched, but not on their search costs, which are sunk by the time the stopping decision is made. Trade the utility intercept against the search cost so as to hold the mean reservation value fixed, and the search *order* is unchanged — but the higher intercept raises realized utility and so makes stopping more likely.

His Appendix D sharpens this into an observable moment: the **recall rate**, the probability of purchasing an alternative encountered earlier in the sequence rather than the most recent one, responds to the utility intercept but not to the search cost. SIFT observes recall directly, because respondents have free recall and we record the entire click sequence. This is our cleanest separation moment, and it is available in the raw data without any modelling.

5.1.3 What the panel buys, and what it does not

Morozov et al. (2021) proves nonparametric identification of individual-level preferences *and* search costs from panel search data. It is worth being precise about the strength of that result, because it is easy to overclaim. The argument assumes the number of sessions per consumer grows without bound; with between 2 and 35 sessions per consumer and a mean of 5, Morozov et al. (2021) estimates hyperparameters rather than individual parameters (his §5.1).

At the task counts SIFT can realistically field — call it $T = 10$ — we are in the same position, and we say so plainly: what the panel delivers is *hierarchical posteriors in the conjoint tradition*, not nonparametric individual identification. That is the standard the conjoint audience already applies to part-worths, and it is the right standard here.

The panel also does something less obvious. Ursu (2018) reports baseline search costs above $102 and attributes the magnitude to observing a single click per search impression with no way to link a consumer's impressions; restricting attention to impressions with two or more clicks cuts mean estimated search costs by up to 83%. Single-spell data mechanically inflates search costs. The panel is therefore a *levels* argument as much as a heterogeneity argument.

5.1.4 What neither the panel nor the shifter buys alone

The two ingredients are complements, and the literature illustrates this by having one or the other but never both. Morozov et al. (2021) has the panel and still normalizes the post-search standard deviation, reporting that little variation in his data speaks to it (his Appendix C). Yavorsky et al. (2021) has an exogenous search-cost shifter and no panel, and does estimate it — finding $\tilde{\sigma} = 8.16$ against the conventional normalization of one, with a confidence interval that excludes one and consequences large enough to reverse the sign of one search-cost coefficient and to understate a counterfactual by a factor of three to four.

SIFT has both. It also has something neither has: because we design the product page, part of the post-search component is *fixed by the design* rather than inferred.

This last point is stronger than it first appears, and it is worth stating carefully. In a field setting the analyst never observes what inspection revealed; the post-search shock is a residual, and all that can be recovered is its variance. In SIFT the analyst *wrote* the product page. So for every clicked alternative the revealed attributes $\mathbf{L}_{ijt}$ are known, and they enter realized utility as $\mathbf{L}'_{ijt}\boldsymbol{\kappa}$ — a mean shift, not a variance. Preferences over attributes that are only visible *after* a click are therefore identified in exactly the way conjoint part-worths are: from which alternative gets bought given what was shown.

The distinction is easy to lose in implementation, with consequences that are silent. The ex ante spread $\tilde{\sigma}$ belongs in the reservation value, because that is computed before the click; the realized $\mathbf{L}$ belongs in the utility, because that is evaluated after it. Use $\tilde{\sigma}$ for both and $\boldsymbol{\kappa}$ enters the likelihood only through $\tilde{\sigma}^2 = \boldsymbol{\kappa}'\mathrm{Var}(\mathbf{L})\boldsymbol{\kappa} + \sigma_\xi^2$, where it trades off exactly against σ_ξ and is not identified at all — while every other parameter continues to recover normally, so nothing looks wrong. This is the sharpest version of the identification claim in the paper and it belongs in the introduction.

5.2 Normalization

Setting the standard deviation of the pre-search shock to one fixes the scale of utility, as in 2 §4.4.2 and Morozov et al. (2021) Appendix C. That normalization is innocuous and we adopt it.

The post-search standard deviation $\tilde{\sigma}$ is a different matter. It is identified in principle by functional form — $\tilde{\sigma}$ appears on both sides of $g(\zeta) = c/\tilde{\sigma}$ — but Yavorsky et al. (2021) shows by simulation that functional-form identification alone is not enough to estimate it in practice, and that an exogenous, alternative-specific cost shifter is what makes estimation work.

The consequence of fixing it deserves to be stated without hedging. With both variances normalized, search costs cannot be monetized and search-cost counterfactuals are not interpretable (2 §4.4.4). That would rule out every friction counterfactual in Section 8. Estimating $\tilde{\sigma}$ is therefore load-bearing for this paper rather than a refinement, and it is the single clearest reason SIFT needs a randomized position rather than merely a rich attribute design.

5.3 Testing the Zero-Search-Cost Restriction

SIFT nests conjoint: as the search cost goes to zero every alternative is inspected, choice is made over fully realized utilities, and the model collapses to a standard choice-based conjoint model. Whether search costs are distinguishable from zero is therefore a testable question rather than a maintained assumption — but the test requires care.

> ⚠ The obvious test is invalid
>
> The natural move is a Wald test on the estimated search cost. It does not work. Zero is a limit rather than an interior point of the parameter space: $c \to 0$ corresponds to $g(c) \to +\infty$ and $\log c \to -\infty$. The null sits at a boundary under every parameterization, so the standard asymptotics do not apply.

We construct the test from observable implications instead, in three ways. The first is holdout fit: SIFT against the nested conjoint model on tasks withheld from estimation. The second is the observable implication itself — whether any respondent leaves alternatives unopened, which under zero search costs no one would. The third is a likelihood-ratio comparison against a *simulated* rather than χ^2 reference distribution.

This also bears on a parameterization choice. 2 §5.2 notes that estimating g directly avoids a root-find inside the optimization loop. Convenient — but it puts the quantity we most want to test on a scale where the null is at infinity. We estimate $\log c$ and pay the root-find, which in our implementation is a spline evaluation rather than an iterative solve (see R/README.md).

5.4 Individual-Level Estimation from a Task Panel

There are two routes to the individual-level deliverable that makes SIFT a conjoint substitute rather than a search paper.

Full hierarchical Bayes is what the conjoint audience expects and what makes individual part-worths and individual search costs directly reportable.

Two-stage empirical Bayes estimates the hyperparameters by simulated maximum likelihood and then draws individual posteriors by Metropolis–Hastings conditional on those estimates. This is what Morozov et al. (2021) does to compute personalized prices (his §7.2 and Appendix H), and it is considerably cheaper.

Our runtime work reverses the ordering we originally expected. Because HB never integrates over the heterogeneity distribution — each respondent sits at their own $\boldsymbol{\theta}_i$ — the mixing-draw multiplier that makes random-coefficients SMLE expensive simply does not arise. On measured per-evaluation costs, full HB is the *cheaper* of the two routes at SIFT's dimensions, not the fallback. We therefore treat HB as the primary estimator and empirical Bayes as the robustness check.

5.5 Estimation

Runtime is a binding practical constraint rather than an implementation detail: a fit has to complete overnight, not over a week. The published timings make the point. 2 Table 2, for 1,000 consumers with heterogeneity, report 5h20m for importance sampling, 29h48m for the kernel-smoothed simulator plus roughly 32 further hours to calibrate its scaling vector, and 198h24m for GHK. Chung et al. (2024) Table 1, for a cross-sectional design of comparable size, reports 0.48 minutes for their probability-mapping simulator against 5–7 minutes kernel-smoothed and 15–127 minutes for crude frequency — one to two orders of magnitude.

We have ported and validated the Chung et al. (2024) estimator and reproduced their Table 1; the port, its five validation checks, and two bugs we found in their shipped code are documented in R/README.md.

For SIFT itself we exploit a structural feature of the design. Conditional on the observed purchase, every Weitzman condition is a conjunction of linear inequalities — the stopping rule is a statement about a maximum and therefore a union in general, but the choice rule names which alternative attains that maximum. Continuation then collapses to the single binding constraint z_{s_K}, because reservation values decrease along the search order. And conditional on $\boldsymbol{\eta}$ and on the post-search shock to the chosen alternative, the remaining shocks are independent one-sided constraints that integrate out in closed form, leaving a one-dimensional integral.

This yields a hybrid estimator: GHK on the pre-search block, quadrature on the post-search block. Measured against the kernel-smoothed accept–reject simulator of Yavorsky et al. (2021) at equal draw counts, its simulation variance is lower by a factor of 21 to 100 depending on the design. Because runtime is exactly linear in the number of draws, the relevant summary is time to a given precision, on which the hybrid is 3 to 21 times cheaper; crude frequency is not competitive on either margin. The kernel-smoothed simulator also shows the upward bias in the negative log-likelihood that Yavorsky et al. (2021) notes is intrinsic to it. The construction and its validation are in `notebooks/sift_likelihood.qmd`.

5.6 Software

All estimation code is in R and is released with the paper. The reservation-value inverse g^{-1} is evaluated by a cubic spline built once in the standardized argument $c/\tilde{\sigma}$, so a single spline serves every $\tilde{\sigma}$; it covers costs down to 3.9×10^{-16}, well below the range any optimizer will propose.

6 Simulation Study

6.1 Design

6.2 Parameter Recovery

6.3 Power to Detect a Nonzero Search Cost

6.4 Misspecification and Behavioral Robustness

7 Empirical Application

7.1 Data and Design

7.2 Descriptive Evidence

7.3 Estimates

7.4 Convergent Validity against Conjoint

7.5 Holdout Prediction

7.6 Task-Order Effects

8 What the Model Delivers

8.1 Decomposing Non-Purchase

8.2 Position and Merchandising Counterfactuals

8.3 Information Architecture

8.4 Search-Based Segmentation

9 Discussion

9.1 What the Nesting Result Changes

9.2 Implications for Practice

9.3 Limitations

9.4 Where Conjoint Remains the Right Tool

10 Conclusion

11 Appendix

11.1 Notation Crosswalk

from either literature and to keep our own implementation honest against the two reference codebases.

Table 1: Notation crosswalk

Object	This paper	2	Yavorsky et al. (2021)
Utility	u_{itj}	u_{ij}	u_{ij}
Pre-search utility	δ_{itj}	δ_{ij}	δ_{ij}
Observed pre-search utility	$\xi_{itj} = \mathbf{X}'_{itj}\boldsymbol{\beta}_i$	$\xi_{ij} = X'_j\boldsymbol{\beta}_i - \alpha_i p_{ij}$	$x'_j\boldsymbol{\beta}$
Pre-search taste shock	μ_{itj}	μ_{ij}	η_{ij}
Post-search taste shock	ε_{itj}	ε_{ij}	ε_{ij}
Observed post-search utility	$\mathbf{L}'_{itj}\boldsymbol{\kappa}_i$	$L'_j\kappa_i$ (§2.4 only)	—
Post-search s.d.	σ_ε	σ_ε	σ (“MVSD”)
Search cost	c_{itj}	$c_{ij} = Z'_j\gamma_i$	$c_{ij} = \exp\{\gamma_0 + d'_{ij}\gamma\}$
Reservation value	z_{itj}	z_{ij}	z_{ij}
Reservation offset	$g(c)$	$g(c)$, $m(c/\sigma_\varepsilon)$	ζ_{ij}
Searched set	$\mathcal{S}_{it}$	S_i	(implicit)
Unsearched set	$\bar{\mathcal{S}}_{it}$	$\bar{S}_i$	(implicit)
Number of searches	H_{it}	H_i	K_i
Search-order index	h	h	(folded into j)
Purchase	y_{it}	y_i	j^*
Parameter vector	$\boldsymbol{\theta}_i$	θ	— ($\theta = \log\sigma$)
Population parameters	$\boldsymbol{\Omega}$	Ω	—
Smoothing constants	ρ_k	ρ_k	λ_k

Four collisions are worth naming explicitly, because each is a live source of error when porting code or reading the two papers side by side.

1. **The differenced conditions ν_1 and ν_2 are swapped between the two papers.** 2 label ν_{1h} the selection (order) condition and ν_{2h} the continuation condition; Yavorsky et al. (2021) label ν_1 continuation and ν_2 selection. Smoothing constants therefore do not correspond element-wise across the two implementations. We follow

   2.

2. θ **denotes different objects.** 2 use it for the full parameter vector; Yavorsky et al. (2021) use it specifically for $\log\sigma$. We follow the former.

3. ζ **and** ξ **are visually near-identical in print** and denote unrelated objects across the two papers — the reservation-value offset in one, observed pre-search utility in the other. We retain ξ and write the offset as $g(\cdot)$.
4. d **is a simulation-draw index in one and a distance vector in the other.** We use it for neither.

11.2 Additional Appendices

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